Sentinel: In-Database Real-Time Fraud & Anomaly Engine

Course: CS/AI/AIM/DS 202 Database Management Systems (Semester III, A.Y. 2026-27)
Institution: ICFAI Foundation for Higher Education (IcfaiTech), Hyderabad **Authors:** Sai Haresh Anand S (25STUCHH010419) and Dulam Seshivardhini (25STUCHH001048)
Platform: MySQL 8.4 (developed in Docker), plain SQL, no application code in the detection path

---

1. Abstract

Most fraud checks live in the application that accepts the payment. The trouble is that anything which writes to the database without going through that application (a batch job, an admin script, a second service) creates transactions that nobody scored. Sentinel puts the check inside the database instead. Every `INSERT` into `transactions` fires an `AFTER INSERT` trigger, which calls the stored procedure `score_transaction()`. The procedure applies six rules: amount Z-score, impossible travel (Haversine distance), frequency velocity, unseen device, off-hours activity and merchant risk. The thresholds come from the `rule_config` table. If any rule fires, a row goes into `alerts` with a JSON array listing the reasons.

The database has 13 tables in third normal form, 11 views, one user-defined function, four procedures, three triggers, a nightly event and four roles. It is seeded with 12,067 transactions across 500 users and 597 accounts. Fund transfers go through `transfer_funds()`, which takes row locks in a fixed order so two opposite transfers cannot deadlock. On a laptop, the trigger takes an insert from about 0.04 ms to about 2 ms, roughly 500 scored transactions per second. A date-range report dropped from a full scan (cost 1,349) to an index range scan (cost 174) once `txn_time` was indexed. 102 SQL assertions and 27 Python integration tests check all of it.

2. Problem Statement and Objectives

2.1 Why application-tier fraud checks fall short

An application-side check has five problems.

1. It only runs where a developer remembered to call it. Direct SQL, data imports and a second service all skip it.
2. If the application commits first and scores afterwards, there is a gap in which an unscored transaction is visible and can be spent.
3. Scoring needs history: the account's mean and deviation, its previous location, the last five minutes of activity. Fetching that over the network for every payment costs latency, when the database already has it sitting next to an index.

4. Thresholds are hard-coded, so changing one means a redeploy.
5. The reason an alert fired usually ends up in a log line instead of a table you can query.

2.2 Objectives

#	Objective	How Sentinel meets it
O1	Score every transaction regardless of who inserted it	`AFTER INSERT` trigger on `transactions`
O2	Explain every alert	`alerts.reasons` JSON array, one entry per rule fired
O3	Make policy data, not code	`rule_config` holds every threshold and an enabled flag
O4	Keep money movement all-or-nothing	`transfer_funds()` with `START TRANSACTION`, `FOR UPDATE`, `COMMIT/ROLLBACK`
O5	Least-privilege access	Four roles; the reporting role cannot read raw transactions
O6	Measure the cost honestly	`EXPLAIN ANALYZE` plans and timed inserts, with limitations documented

3. Requirements and Entity-Relationship Design

3.1 Functional requirements

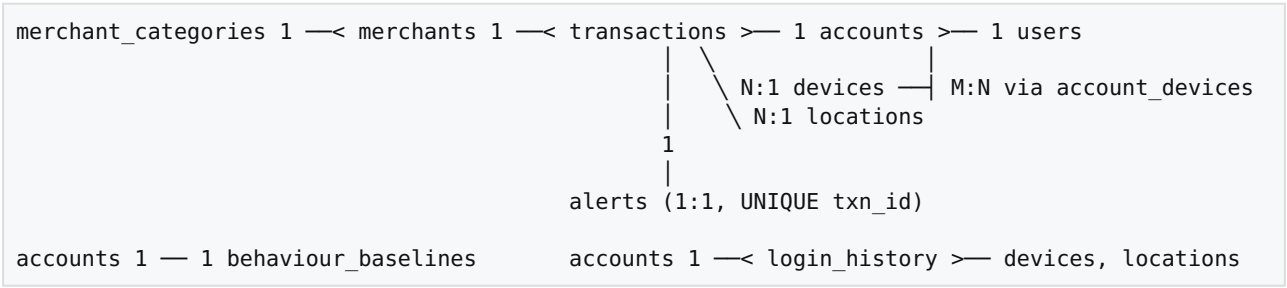
- Store users, their accounts, the devices and locations they transact from, and the merchants they pay.
- Record each transaction with amount, time, merchant, device, location and status.
- On each new transaction, decide whether it is suspicious and record why.
- Let an analyst move an alert through `open → investigated → confirmed | false_positive`, with the review time stamped and the change audited.
- Provide reporting views for dashboards without exposing base tables.
- Move funds between accounts atomically.

3.2 Entity sets

Table	Role	Primary key
`merchant_categories`	Lookup: category names	`category_id`
`locations`	Lookup: city, state, country, latitude, longitude	`location_id`
`devices`	Device type, OS, browser, fingerprint hash	`device_id`
`rule_config`	One row per fraud rule: enabled flag, threshold, parameters JSON	`rule_id`
`users`	Customer identity (name, unique email, phone)	`user_id`
`accounts`	Savings or current account owned by a user, with balance and status	`account_id`
`merchants`	Merchant with category, home coordinates, risk score	`merchant_id`
`account_devices`	Associative table for the many-to-many link between accounts and devices	(`account_id`, `device_id`)
`transactions`	The fact table; 12,067 rows	`txn_id`

Table	Role	Primary key
`alerts`	Output of the scoring engine	`alert_id`
`behaviour_baselines`	Per-account average, deviation, usual hours and cities	`baseline_id`
`login_history`	Account logins by device and location	`login_id`
`audit_log`	Who changed which row, with old and new values as JSON	`audit_id`

3.3 Relationships and foreign-key mapping



Foreign key	References	On delete	Reasoning
`accounts.user_id`	`users`	CASCADE	An account has no meaning without its owner
`merchants.category_id`	`merchant_categories`	RESTRICT	A category in use cannot disappear
`account_devices.account_id` / `device_id`	`accounts` / `devices`	CASCADE	A link row dies with either side
`transactions.account_id`	`accounts`	CASCADE	History belongs to the account
`transactions.merchant_id` / `device_id` / `location_id`	`merchants` / `devices` / `locations`	RESTRICT	Referenced reference data must not vanish silently
`alerts.txn_id`	`transactions`	CASCADE	An alert explains one transaction
`behaviour_baselines.account_id`	`accounts`	CASCADE	Derived data follows its account
`login_history.account_id`	`accounts`	CASCADE	
`login_history.device_id` / `location_id`	`devices` / `locations`	RESTRICT	

Other declared constraints: `UNIQUE` on `users.email`, `devices.fingerprint_hash`, `rule_config.rule_name`, `merchant_categories.category_name`, `(locations.city, locations.country)`, `alerts.txn_id` and `behaviour_baselines.account_id`; `CHECK (amount > 0)` on transactions; `CHECK (balance >= 0)` on accounts; range checks on latitude and longitude.

The schema was built week by week for the lab: Week 2 creates bare tables and Week 3 adds keys and constraints with `ALTER TABLE`. The end result is the same as declaring everything inline, and a fingerprint script compares it against a reference. The alerts table starts as `fraud_alerts` in Week 2 and is renamed to `alerts` later (its unused `notes` column is dropped at the same time), so older week notes may still use the first name.

4. Normalised Relational Schema (1NF, 2NF, 3NF)

For each table we wrote down the functional dependencies (FDs) that follow from the business rules, found the candidate keys, and checked three conditions.

- **1NF:** every column is atomic and single-valued, and every row is identified by a key.
- **2NF:** no non-key attribute depends on a proper part of a composite key.
- **3NF:** for every non-trivial FD X → A, either X is a superkey or A is part of a candidate key. Equivalently, no non-key attribute depends on another non-key attribute.

4.1 Table-by-table verification

Table	Candidate keys	Non-trivial functional dependencies	Result
`users`	`user_id`, `email`	`user_id → full_name, email, phone, created_at`; `email → user_id`	Every determinant is a candidate key → 3NF (BCNF)
`accounts`	`account_id`	`account_id → user_id, account_type, balance, status, created_at`	Single-column key rules out partial dependency (2NF). No non-key column determines another → 3NF
`merchants`	`merchant_id`	`merchant_id → merchant_name, category_id, city, latitude, longitude, risk_score`	The category name is not stored here; it is reached through `category_id`, so `category_id → category_name` stays in `merchant_categories` → 3NF
`transactions`	`txn_id`	`txn_id → account_id, merchant_id, device_id, location_id, amount, txn_time, status, metadata_json`	City, device type and merchant name are not repeated here; they live in the tables the foreign keys point to → 3NF
`rule_config`	`rule_id`, `rule_name`	`rule_id → rule_name, enabled, threshold_value, parameters_json, updated_at`; `rule_name → rule_id`	Both determinants are candidate keys → 3NF (BCNF)
`alerts`	`alert_id`, `txn_id`	`alert_id → txn_id, rule_name, risk_score, reasons, status, created_at, reviewed_at`; `txn_id → alert_id`	Both determinants are candidate keys → 3NF (BCNF)
`account_devices`	(`account_id`, `device_id`)	(`account_id`, `device_id`) → `first_seen`, `last_seen`	The only composite-key table. Both columns describe the pairing, not either side alone, so there is no partial dependency → 2NF and 3NF

The remaining six tables (`merchant_categories`, `locations`, `devices`, `behaviour_baselines`, `login_history`, `audit_log`) each have a single-column surrogate key and no dependency between non-key columns. In `locations`, `city → country` is not declared as a dependency because the same city name can occur in more than one country; the candidate key is (`city`, `country`) and `latitude`, `longitude` depend on it.

4.2 1NF in the presence of JSON columns

`alerts.reasons`, `rule_config.parameters_json`, `transactions.metadata_json` and the two JSON columns in `behaviour_baselines` hold structured documents. We treat them as atomic values, because the engine never joins on or filters by anything inside them. `reasons` is output that an analyst reads whole, and `parameters_json` holds free-form rule parameters. If we ever needed "all alerts that cite rule X", that would be the point to split out a child table.

4.3 One deliberate redundancy

`behaviour_baselines` stores values that could be computed from `transactions` (mean, standard deviation, active hours, typical cities). We kept it as a cache. The nightly procedure rebuilds it in full, so it is never more than a day stale, and reports do not have to aggregate an account's whole history each time.

5. Advanced Database Features

5.1 The six fraud rules

Each rule that fires adds one point to the score and appends a readable string to the `reasons` array. One fired rule is enough to raise an alert. There is no combined threshold; each rule has its own, read from `rule_config` when the transaction is scored.

#	Rule	Condition	Default threshold
1	Amount Z-score	`(amount − AVG) / STDDEV_SAMP` over the account's history exceeds the threshold; requires at least 5 prior-or-current rows and a non-zero deviation	3
2	Travel velocity	Haversine distance to the previous transaction is above 50 km and implied speed exceeds the limit	900 km/h
3	Frequency velocity	Number of transactions by the account inside the window, including the new one, exceeds the limit	more than 5 in 5 minutes
4	New device	The device is not in `account_devices` for this account	1 (enabled)
5	Off-hours	The hour of day accounts for less than 2% of the account's history	0.02
6	Merchant risk	The merchant has at least 3 transactions and at least 3 alerts, and its alert rate divided by the platform-wide alert rate exceeds the threshold	2 (min. alerts 3)

Haversine is implemented as the deterministic function `haversine_km(lat1, lon1, lat2, lon2)` with Earth radius R = 6371 km:

```
a = sin²(Δφ/2) + cos φ1 · cos φ2 · sin²(Δλ/2)
d = 2R · asin(√a)
```

Hyderabad (17.385, 78.4867) to Tokyo (35.6762, 139.6503) comes out at 6,311.39 km in MySQL. If the two payments are 15 minutes apart, the implied speed is 25,246 km/h against a 900 km/h limit. The 50 km floor is there because two nearby places with close

timestamps would otherwise give a huge speed from a tiny distance divided by a tiny time.

The Z-score rule has a ceiling. The trigger fires after the insert, so the new row is already part of the average and standard deviation it is compared with. With n values, one outlier can never score above $(n - 1)/\sqrt{n}$, which is about 7.9 for 65 transactions. That makes the rule a little conservative on short histories. We left it and listed it as a limitation, because using only earlier rows would cost a second query on every insert.

5.2 Scoring pipeline: `trg_after_txn_insert` and `score_transaction()`

```
CREATE TRIGGER trg_after_txn_insert
AFTER INSERT ON transactions
FOR EACH ROW
BEGIN
    CALL score_transaction(NEW.txn_id);
END;
```

The trigger is `AFTER` because the procedure reads the stored row, and a `BEFORE` trigger runs before that row exists. It is a row-level trigger, so a multi-row `INSERT` scores each row.

`score_transaction(p_txn_id)` loads the transaction and the `rule_config` rows, evaluates each enabled rule, builds the `reasons` array with `JSON_ARRAY`, and, if the score is above zero, writes

```
INSERT INTO alerts (txn_id, rule_name, risk_score, reasons, status)
VALUES (..., 'combined', v_score, v_reasons, 'open')
ON DUPLICATE KEY UPDATE ...;
```

Because `txn_id` is `UNIQUE`, scoring the same transaction twice updates the alert instead of duplicating it. If the score comes out as zero, any open alert for that transaction is deleted, so re-running after a threshold change keeps the table correct. `backfill_alerts(first, last)` applies the same logic to a range of old transactions.

Two more triggers cover the analyst workflow. `trg_before_alert_update` stamps `reviewed_at` when an alert leaves `open`. `trg_after_alert_update` writes the old and new status to `audit_log`. It uses `USER()` and not `CURRENT_USER()`, because inside a trigger `CURRENT_USER()` returns the definer, and every audit row would then say root.

5.3 Baseline maintenance: `refresh_behaviour_baselines()`

```
CREATE PROCEDURE refresh_behaviour_baselines()
BEGIN
    DECLARE EXIT HANDLER FOR SQLEXCEPTION BEGIN ROLLBACK; RESIGNAL; END;
    START TRANSACTION;
    DELETE FROM behaviour_baselines;
    INSERT INTO behaviour_baselines (account_id, avg_amount, stddev_amount,
                                     peak_hours_json, typical_cities_json)
    SELECT ... AVG(amount), STDDEV_SAMP(amount), JSON_ARRAYAGG(hour), JSON_ARRAYAGG(city)
    FROM transactions ... GROUP BY account_id;
    COMMIT;
END;
```

The delete and rebuild share one transaction with an exit handler, so if anything fails the old baselines stay and the table is not left empty. The event `evt_refresh_baselines` calls it daily at 02:30, which needs the event scheduler to be on.

5.4 ACID transfers: `transfer_funds()`

`transfer_funds()` moves money between two accounts and records the debit as a transaction, so the scoring trigger evaluates it like any other payment.

```
DECLARE EXIT HANDLER FOR SQLEXCEPTION BEGIN ROLLBACK; RESIGNAL; END;

IF p_amount IS NULL OR p_amount <= 0 THEN SIGNAL SQLSTATE '45000' ...;
IF p_from_account = p_to_account      THEN SIGNAL SQLSTATE '45000' ...;

START TRANSACTION;

IF p_from_account < p_to_account THEN
    SELECT balance, status INTO v_from_balance, v_from_status
      FROM accounts WHERE account_id = p_from_account FOR UPDATE;
    SELECT status INTO v_to_status
      FROM accounts WHERE account_id = p_to_account   FOR UPDATE;
ELSE
    SELECT status INTO v_to_status
      FROM accounts WHERE account_id = p_to_account   FOR UPDATE;
    SELECT balance, status INTO v_from_balance, v_from_status
      FROM accounts WHERE account_id = p_from_account FOR UPDATE;
END IF;

-- missing / inactive account or insufficient funds: ROLLBACK, then SIGNAL

UPDATE accounts SET balance = balance - p_amount WHERE account_id = p_from_account;
UPDATE accounts SET balance = balance + p_amount WHERE account_id = p_to_account;
INSERT INTO transactions (account_id, merchant_id, device_id, location_id, amount, status)
VALUES (p_from_account, 1, p_device_id, p_location_id, p_amount, 'success');

COMMIT;
```

Property	Mechanism
Atomicity	Debit, credit and the transaction record commit together; the exit handler rolls back on any error, including one raised inside the scoring trigger
Consistency	Validation in the procedure, plus `CHECK (balance >= 0)` as a backstop
Isolation	`SELECT … FOR UPDATE` takes exclusive row locks, so a concurrent transfer on the same account waits instead of reading a stale balance
Durability	InnoDB redo log flushed at `COMMIT`

Deadlock is the main risk. If a transfer A→B and a transfer B→A each locked their own source account first, each would then wait on the other forever. The procedure therefore always locks the lower `account_id` first. Every transfer takes locks in the same order, so a cycle of waits cannot form.

Only the sender's debit is recorded as a transaction, against placeholder merchant 1. Money received is not a purchase, and recording it would distort the receiver's spending statistics.

5.5 Indexing and performance

The indexes were added in Week 9, after the schema and the rule queries existed, so each one serves a query we had already written:

Index	Serves
`idx_txn_account_time (account_id, txn_time)`	Per-account velocity, previous-transaction lookup, baselines, latest-N queries
`idx_txn_merchant_time (merchant_id, txn_time)`	Merchant risk and merchant timelines
`idx_txn_time (txn_time)`	Date-range reports
`idx_txn_location (location_id)`	Location joins
`idx_alert_status (status, created_at)`	The open-alerts view
`idx_alert_rule (rule_name)`	Alerts by rule

Measured with `EXPLAIN ANALYZE` on MySQL 8.4.11:

Query	Before	After	Plan change
Daily report over a 6-day range (reads of 12,067 rows → 315)	cost 1,349, table scan	cost 174, range scan on `idx_txn_time`	about 8× cheaper
Latest 50 transactions of one account	cost 18.2 with filesort	cost 18.2, no sort	Same cost; the composite index removes the `Sort` step because rows come out already ordered
Merchant 50, newest 20	cost 88.5	reverse scan on `idx_txn_merchant_time`	no sort

So the drop from 1,349 to 174 comes from the single-column index on `txn_time` serving a range predicate. The composite index on `(account_id, txn_time)` does something different. An account has only about 52 rows on average, so the optimiser's cost barely moves, but the velocity and travel rules get an account's recent rows already in time order, with no sort. We report both numbers as measured and do not give the composite index credit for the bigger one.

6. Experimental Benchmarks and Evaluation

All measurements used MySQL 8.4.11 in Docker on a laptop under WSL2, one connection, with the 12,067-transaction data set (500 users, 597 accounts, 50 merchants).

6.1 Trigger overhead

1,000 inserts were timed inside a transaction that was then rolled back, once with the trigger dropped and once with it present.

Configuration	Time for 1,000 inserts	Per row
No trigger	about 41 ms	about 0.04 ms

Configuration	Time for 1,000 inserts	Per row
With `trg_after_txn_insert`	about 1,980 ms	about 2 ms

Scoring makes each insert about 50 times slower, which is about 2 ms in absolute terms, or roughly 500 scored transactions per second on one connection. If a person is waiting on a card or UPI confirmation, 2 ms does not matter. For bulk loading it does.

6.2 Bulk loading

When we loaded the seed data with the trigger already installed, it stalled at 5,925 rows after 250 seconds, since every row ran the full scoring procedure against a growing table. So the repository loads the data first and creates the trigger last (about 20 s for weeks 1–9), then scores the history in one pass with `backfill_alerts(1, 12067)`. That takes about 38 s and produces 634 alerts, about 5% of transactions.

6.3 Detection check

We planted 27 persona transactions (ids 12041–12067) to exercise each rule. They produce 21 alerts, and an ordinary purchase produces none. The 102 SQL assertions across 11 weeks and the 27 Python tests all pass as of this writing.

6.4 Known limitations

- The seed data assigns random cities to transactions, so the travel rule fires 32–46 times on data that is synthetic rather than fraudulent.
- On amounts with a skewed distribution, a Z-score above 3 flags about 1.2% of rows (145 of 12,067). Amount is a poor fit for a normal-distribution test; a log transform would help.
- The off-hours rule is unreliable below about 50 transactions per account.
- Merchant risk feeds on alerts that other rules produce, so it can reinforce itself. The `min_alerts = 3` guard limits this but does not remove it.
- The baseline includes the row being scored (see 5.1).
- There is no user interface of its own (`demo.html` is only an explainer), and the code uses MySQL-specific syntax.

7. Viva Questions and Discussion

Q1. What does the scoring trigger cost, and is that acceptable? About 2 ms per inserted row in our test, against 0.04 ms without it, so one connection scores around 500 transactions per second. For an interactive payment that is well under anything a user would notice, and the work is a handful of indexed lookups. Bulk loading is where it hurts: our seed load stalled with the trigger in place, so we load first, create the trigger afterwards and use `backfill_alerts()` for history. Beyond a few thousand scored inserts per second we would move scoring to an asynchronous queue, and give up the guarantee that scoring finishes before the insert returns.

Q2. Why `AFTER INSERT` and not `BEFORE`? `score_transaction()` looks the transaction up by id and aggregates over the account's rows including this one, so the row must already exist. A `BEFORE` trigger can change `NEW` or reject the row, but it runs before the row is stored. We use `BEFORE UPDATE` only where changing `NEW` is the whole point, to stamp `reviewed_at`. The price of `AFTER` is that the new row is part of the average and deviation, which caps the Z-score near $(n - 1)/\sqrt{n}$.

Q3. Explain the locking in `transfer_funds()`. How is deadlock avoided? `SELECT … FOR UPDATE` inside a transaction takes an exclusive InnoDB row lock on the account and holds it until `COMMIT` or `ROLLBACK`. A second transfer on the same account waits, and when it resumes it reads the committed balance, so two transfers cannot spend the same money. A deadlock needs two sessions that each hold one lock and want the other's. We always lock the lower account id first, so A→B and B→A both start with the same row and the second one just waits. Without that rule InnoDB would detect the cycle and abort one of them. With it, that situation never arises.

Q4. What is the trade-off of putting fraud logic in the database? In its favour: no writer can bypass it, there is no gap between commit and check, history is read next to its indexes, and an administrator can change a threshold without a release. Against it: the scoring uses database CPU on the write path, the code is MySQL-specific and harder to test and debug than application code, a bug in a trigger can make the original insert fail, and the database is usually the hardest tier to scale, which now carries scoring as well. We handle this by keeping the rules simple, indexed and set-based, and by keeping any model-style analysis outside the database.

Q5. The report shows cost falling from 1,349 to 174. What caused it, and what does the composite index do? The drop is a date-range report on `transactions` going from a full table scan of 12,067 rows to an index range scan of 315 rows, using the single-column index `idx_txn_time`. The composite index `(account_id, txn_time)` follows the leftmost-prefix rule: it helps filters on `account_id`, or `account_id` plus a time range, and returns an account's rows already sorted by time. For "latest 50 of an account" it did not lower the cost (18.2 either way, since an account has only about 52 rows), but it removed the sort step, which the scoring rules benefit from. It does nothing for a filter on `txn_time` alone, which is why we have both indexes.

8. Conclusion and Future Scope

Sentinel shows that a working fraud check can sit entirely inside a relational database. One trigger makes sure every transaction is scored, whoever inserted it. Thresholds are ordinary rows, every alert carries a JSON explanation, and transfers stay atomic under concurrent access because locks are always taken in the same order. Scoring costs about 2 ms per insert, and indexing brought a representative report from cost 1,349 down to 174. The evaluation also records the weak spots: the seeded cities are random, so the travel rule fires on synthetic data; a Z-score suits skewed amounts poorly; and the baseline includes the row being tested.

Future work, roughly in the order we would do it:

1. Compute the baseline from earlier transactions only, which removes the Z-score ceiling.
2. Give each account a home city, so the travel rule compares against habitual locations and not only the previous row.
3. Base merchant risk on confirmed fraud (`alerts.status = 'confirmed'`) instead of raw alerts, to break the feedback loop.
4. Replace the plain Z-score with a robust one (median and MAD, or a Z-score on log amounts).
5. Put a thin read-only API on the `api_reader` role, with a dashboard over the existing reporting views.
6. Add a `txn_type` column so transfers, purchases and refunds can be scored differently.
7. Try the asynchronous alternative (queue plus worker) and measure its latency and throughput against the in-trigger design.

Figures in this report were measured on the project database; the supporting scripts and EXPLAIN output are in `week-09-views-indexes-dcl/docs/benchmark_measured.md` and `week-11-final-polish/docs/performance.md`.

Appendix A. Code Excerpts

Only the pieces discussed in the report are reproduced here, with the explanatory comments removed to save space. The complete, commented scripts are in the repository under `week-NN-*/sql/`.

A.1 Indexes

`week-09-views-indexes-dcl/sql/30_indexes.sql`

```
USE sentinel;

CREATE INDEX idx_txn_account_time  ON transactions (account_id, txn_time);
CREATE INDEX idx_txn_merchant_time ON transactions (merchant_id, txn_time);
CREATE INDEX idx_txn_location      ON transactions (location_id);
CREATE INDEX idx_txn_time          ON transactions (txn_time);

CREATE INDEX idx_alert_status ON alerts (status, created_at);
CREATE INDEX idx_alert_rule   ON alerts (rule_name);

CREATE INDEX idx_lh_account_time   ON login_history (account_id, login_time);
CREATE INDEX idx_audit_table_time  ON audit_log (table_name, changed_at);

CREATE INDEX idx_ad_device ON account_devices (device_id);
```

A.2 Triggers

`week-10-procedures-triggers/sql/30_triggers.sql`

```
USE sentinel;
```

```
DROP TRIGGER IF EXISTS trg_after_txn_insert;
DELIMITER //
CREATE TRIGGER trg_after_txn_insert
AFTER INSERT ON transactions
FOR EACH ROW
BEGIN
    CALL score_transaction(NEW.txn_id);
END //
DELIMITER ;

DROP TRIGGER IF EXISTS trg_before_alert_update;
DELIMITER //
CREATE TRIGGER trg_before_alert_update
BEFORE UPDATE ON alerts
FOR EACH ROW
BEGIN
    IF OLD.status = 'open' AND NEW.status <> 'open' AND NEW.reviewed_at IS NULL THEN
        SET NEW.reviewed_at = CURRENT_TIMESTAMP;
    END IF;
END //
DELIMITER ;

DROP TRIGGER IF EXISTS trg_after_alert_update;
DELIMITER //
CREATE TRIGGER trg_after_alert_update
AFTER UPDATE ON alerts
FOR EACH ROW
BEGIN
    IF OLD.status <> NEW.status THEN
        INSERT INTO audit_log (table_name, record_id, action, old_value_json, new_value_json,
changed_by)
        VALUES ('alerts', NEW.alert_id, 'UPDATE',
                JSON_OBJECT('status', OLD.status),
                JSON_OBJECT('status', NEW.status),
                USER());
    END IF;
END //
DELIMITER ;
```

A.3 The `api_reader` role

`week-09-views-indexes-dcl/sql/40_roles_dcl.sql`

```
CREATE ROLE IF NOT EXISTS 'api_reader';
GRANT SELECT ON sentinel.v_active_alerts      TO 'api_reader';
GRANT SELECT ON sentinel.v_daily_txn_summary  TO 'api_reader';
GRANT SELECT ON sentinel.v_top_risky_users    TO 'api_reader';
GRANT SELECT ON sentinel.v_alerts_by_city_day TO 'api_reader';
GRANT SELECT ON sentinel.rule_config          TO 'api_reader';
GRANT 'api_reader'    TO 'api_user'@'%';
```

The six rule views, `score_transaction`, `transfer_funds` and the constraint definitions are in `week-09-views-indexes-dcl/sql/10_rule_views.sql`, `week-10-procedures-triggers/sql/20_score_transaction.sql`, `50_transfer_funds.sql` and `week-03-constraints/sql/01_constraints.sql`.

Appendix B. Captured Terminal Output

Each figure was produced by running the shown statement against the live `sentinel-db` container (MySQL 8.4) and capturing the unedited result on 21 Oct 2026. Figures B.11 and B.12 are the repository test runs.

```
mysql> SHOW FULL TABLES;

+------------------------+------------+
| Tables_in_sentinel     | Table_type |
+------------------------+------------+
| account_devices        | BASE TABLE |
| accounts               | BASE TABLE |
| alerts                 | BASE TABLE |
| audit_log              | BASE TABLE |
| behaviour_baselines    | BASE TABLE |
| devices                | BASE TABLE |
| locations              | BASE TABLE |
| login_history          | BASE TABLE |
| merchant_categories    | BASE TABLE |
| merchants              | BASE TABLE |
| rule_config            | BASE TABLE |
| rule_impossible_travel | VIEW       |
| rule_merchant_risk     | VIEW       |
| rule_new_device        | VIEW       |
| rule_off_hours         | VIEW       |
| rule_velocity          | VIEW       |
| rule_zscore            | VIEW       |
| transactions           | BASE TABLE |
| users                  | BASE TABLE |
| v_active_alerts        | VIEW       |
| v_alerts_by_city_day   | VIEW       |
| v_daily_txn_summary    | VIEW       |
| v_top_risky_users      | VIEW       |
| v_transaction_risk     | VIEW       |
+------------------------+------------+
```

Figure B.1 Thirteen base tables and eleven views.

```
mysql> DESCRIBE transactions;

+--------------+----------------------------------+------+-----+-------------------+-------------------+
| Field        | Type                             | Null | Key | Default           | Extra             |
+--------------+----------------------------------+------+-----+-------------------+-------------------+
| txn_id       | bigint                           | NO   | PRI | NULL              | auto_increment    |
| account_id   | int                              | NO   | MUL | NULL              |                   |
| merchant_id  | int                              | NO   | MUL | NULL              |                   |
| device_id    | int                              | NO   | MUL | NULL              |                   |
| location_id  | int                              | NO   | MUL | NULL              |                   |
| amount       | decimal(12,2)                    | NO   |     | NULL              |                   |
| txn_time     | timestamp                        | NO   | MUL | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| status       | enum('success','pending','failed') | NO   |     | success           |                   |
| metadata_json | json                             | YES  |     | NULL              |                   |
+--------------+----------------------------------+------+-----+-------------------+-------------------+
```

Figure B.2 Structure of `transactions`.

```
mysql> SELECT * FROM rule_config;

+---------+-------------------------+---------+-----------------+-----------------------------------------------------------------------------------------------------------------------------------------------------+---------------------+
| rule_id | rule_name               | enabled | threshold_value | parameters_json                                                                                                                                     | updated_at          |
+---------+-------------------------+---------+-----------------+-----------------------------------------------------------------------------------------------------------------------------------------------------+---------------------+
|       1 | zscore                  |       1 |          3.0000 | {"description": "Z-score above which an amount is anomalous"}                                                                                       | 2026-10-21 17:24:44 |
|       2 | off_hours               |       1 |          0.0200 | {"description": "Flag if the account has fewer than 2% of its transactions in this hour"}                                                           | 2026-10-21 16:38:37 |
|       3 | new_device              |       1 |          1.0000 | {"description": "Flag if the device was never seen on this account"}                                                                                | 2026-10-21 17:24:42 |
|       4 | velocity                |       1 |          5.0000 | {"description": "Max transactions allowed inside the velocity window"}                                                                              | 2026-10-21 16:38:37 |
|       5 | velocity_window         |       1 |          5.0000 | {"description": "Velocity window length in minutes"}                                                                                                | 2026-10-21 16:38:37 |
|       6 | impossible_travel       |       1 |        900.0000 | {"description": "Max plausible speed in km/h between two transactions"}                                                                             | 2026-10-21 16:38:37 |
|       7 | merchant_risk           |       1 |          2.0000 | {"description": "Flag if a merchant alert rate is this many times the platform average"}                                                            | 2026-10-21 16:38:37 |
|       8 | merchant_risk_min_alerts |       1 |          3.0000 | {"description": "A merchant needs at least this many alerts before it can be called risky (one alert proves nothing)"} | 2026-10-21 16:38:37 |
+---------+-------------------------+---------+-----------------+-----------------------------------------------------------------------------------------------------------------------------------------------------+---------------------+
```

Figure B.3 The eight `rule_config` rows.

```
mysql> INSERT INTO transactions (account_id,merchant_id,device_id,location_id,amount,status) VALUES (1,1,1,1,-50,'success');

ERROR 3819 (HY000) at line 1: Check constraint 'chk_txn_amount' is violated.
```

Figure B.4 A `CHECK` constraint rejecting a negative amount.

```
mysql> SHOW GRANTS FOR api_reader@'%';

+-----------------------------------------------------------------------------+
| Grants for api_reader@%                                                     |
+-----------------------------------------------------------------------------+
| GRANT USAGE ON *.* TO `api_reader`@`%`                                      |
| GRANT SELECT ON `sentinel`.`rule_config` TO `api_reader`@`%`                |
| GRANT SELECT ON `sentinel`.`v_active_alerts` TO `api_reader`@`%`            |
| GRANT SELECT ON `sentinel`.`v_alerts_by_city_day` TO `api_reader`@`%`       |
| GRANT SELECT ON `sentinel`.`v_daily_txn_summary` TO `api_reader`@`%`        |
| GRANT SELECT ON `sentinel`.`v_top_risky_users` TO `api_reader`@`%`          |
+-----------------------------------------------------------------------------+
```

Figure B.5 Least-privilege grants of `api_reader`.

```
mysql> SELECT TRIGGER_NAME,EVENT_MANIPULATION ev,EVENT_OBJECT_TABLE tbl,ACTION_TIMING timing FROM information_schema.TRIGGERS WHERE TRIGGER_SCHEMA='sentinel';

+-------------------------+--------+-------------+--------+
| TRIGGER_NAME            | ev     | tbl         | timing |
+-------------------------+--------+-------------+--------+
| trg_before_alert_update | UPDATE | alerts      | BEFORE |
| trg_after_alert_update  | UPDATE | alerts      | AFTER  |
| trg_after_txn_insert    | INSERT | transactions | AFTER  |
+-------------------------+--------+-------------+--------+
```

Figure B.6 Triggers registered in the schema.

```
mysql> SELECT ROUTINE_NAME,ROUTINE_TYPE FROM information_schema.ROUTINES WHERE ROUTINE_SCHEMA='sentinel';

+-----------------------------+--------------+
| ROUTINE_NAME                | ROUTINE_TYPE |
+-----------------------------+--------------+
| haversine_km                | FUNCTION     |
| backfill_alerts             | PROCEDURE    |
| refresh_behaviour_baselines | PROCEDURE    |
| score_transaction           | PROCEDURE    |
| transfer_funds              | PROCEDURE    |
+-----------------------------+--------------+
```

Figure B.7 Stored routines.

```
mysql> SELECT alert_id,txn_id,rule_name,risk_score,status,reasons FROM alerts WHERE txn_id BETWEEN 12041 AND 12067 ORDER BY alert_id LIMIT 6;

+----------+--------+-----------+------------+--------+-----------------------------------------------------+
| alert_id | txn_id | rule_name | risk_score | status | reasons                                             |
+----------+--------+-----------+------------+--------+-----------------------------------------------------+
|        1 |  12042 | combined  |       1.00 | open   | ["Impossible travel (1767.65 km in 10 min)"]        |
|        2 |  12043 | combined  |       1.00 | open   | ["Off-hours (2:00, only 1.92% of history)"]         |
|        3 |  12049 | combined  |       1.00 | open   | ["Velocity spike (6 in 5 min)"]                     |
|        4 |  12050 | combined  |       1.00 | open   | ["Velocity spike (7 in 5 min)"]                     |
|        5 |  12051 | combined  |       1.00 | open   | ["Velocity spike (8 in 5 min)"]                     |
|        6 |  12052 | combined  |       2.00 | open   | ["High amount (z=6.97)", "New device (NEVER_SE...)"] |
+----------+--------+-----------+------------+--------+-----------------------------------------------------+
```

Figure B.8 Alerts raised for the scripted persona transactions.

```
mysql> SELECT * FROM v_top_risky_users LIMIT 5;

+------------+-----------------+--------------------+--------------+-----------------+-----------------+
| account_id | full_name       | email              | total_alerts | avg_risk_score | max_risk_score |
+------------+-----------------+--------------------+--------------+-----------------+-----------------+
|          1 | Tammy Allison   | user1@example.com  |            6 |           1.17 |           2.00 |
|         20 | Anna Henderson  | user18@example.com |            1 |           3.00 |           3.00 |
|         46 | Kevin Pena      | user40@example.com |            1 |           3.00 |           3.00 |
|         77 | Rachael Pearson | user65@example.com |            1 |           3.00 |           3.00 |
|         94 | Gina Beard      | user79@example.com |            1 |           2.00 |           2.00 |
+------------+-----------------+--------------------+--------------+-----------------+-----------------+
```

Figure B.9 `v_top_risky_users`.

```
mysql> EXPLAIN SELECT COUNT(*) FROM transactions WHERE txn_time >= '2026-01-01' AND txn_time < '2026-01-08';

+----+-------------+--------------+------------+-------+-----------------------------------------------------------+-------------+---------+------+------+----------+--------------------------+
| id | select_type | table        | partitions | type  | possible_keys                                             | key         | key_len | ref  | rows | filtered | Extra                    |
+----+-------------+--------------+------------+-------+-----------------------------------------------------------+-------------+---------+------+------+----------+--------------------------+
|  1 | SIMPLE      | transactions | NULL       | range | idx_txn_merchant_time,idx_txn_time,idx_txn_account_time | idx_txn_time | 4       | NULL |  405 |   100.00 | Using where; Using index |
+----+-------------+--------------+------------+-------+-----------------------------------------------------------+-------------+---------+------+------+----------+--------------------------+
```

Figure B.10 `EXPLAIN` of a date-range query using `idx_txn_time`.

```
mysql> START TRANSACTION;
mysql> INSERT INTO transactions (account_id,merchant_id,device_id,location_id,amount,status) VALUES (1,1,1,1,95000,'success');
mysql> SELECT alert_id,txn_id,rule_name,risk_score,status,reasons FROM alerts WHERE txn_id=LAST_INSERT_ID();
mysql> ROLLBACK;

+----------+--------+-----------+------------+--------+-----------------------------------------------------+
| alert_id | txn_id | rule_name | risk_score | status | reasons                                             |
+----------+--------+-----------+------------+--------+-----------------------------------------------------+
|       50 |  12112 | combined  |       2.00 | open   | ["High amount (z=4.97)", "New device (93446f17...)"] |
+----------+--------+-----------+------------+--------+-----------------------------------------------------+
```

Figure B.11 Live trigger demonstration (insert, alert appears, rollback).

```
$ bash scripts/check.sh 11
ok    week 01  check.sql (2 assertions)
ok    week 02  check.sql (6 assertions)
ok    week 03  check.sql (8 assertions)
ok    week 04  check.sql (13 assertions)
ok    week 05  check.sql (5 assertions)
ok    week 06  check.sql (6 assertions)
ok    week 07  check.sql (6 assertions)
ok    week 08  check.sql (7 assertions)
ok    week 09  check.sql (20 assertions)
ok    week 10  check.sql (21 assertions)
ok    week 11  check.sql (8 assertions)
----
passed: 50   failed: 0
```

Figure B.12 SQL assertion suite, per-week results (2+6+8+13+5+6+6+7+20+21+8 = 102 assertions in 50 check files).

```
$ pytest -q
.........................                                [100%]
27 passed in 4.96s
```

Figure B.13 Python test suite.

Appendix C. Interactive Demonstration

`demo.html` is a standalone page that re-implements the six rules in JavaScript so the pipeline can be explained without a database connection. It is an explanatory aid; the authoritative behaviour is the SQL above. Screenshots below were captured from the page.

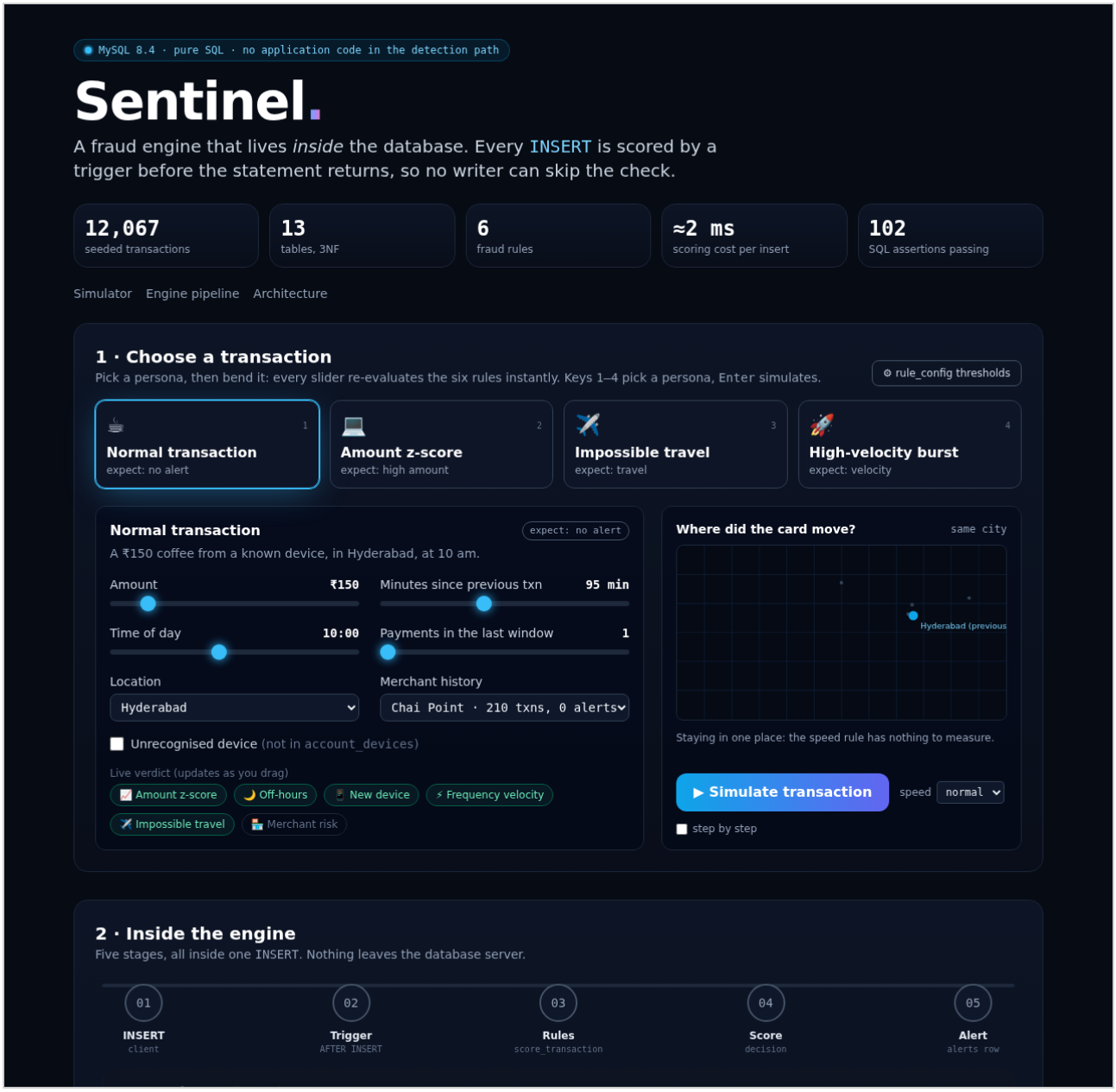


Figure C.1 Landing view.

MySQL 8.4 · pure SQL · no application code in the detection path

Sentinel.

A fraud engine that lives *inside* the database. Every `INSERT` is scored by a trigger before the statement returns, so no writer can skip the check.

12,067	13	6	≈2 ms	102
seeded transactions	tables, 3NF	fraud rules	scoring cost per insert	SQL assertions passing

Simulator · Engine pipeline · Architecture

1 · Choose a transaction

Pick a persona, then bend it: every slider re-evaluates the six rules instantly. Keys 1–4 pick a persona, Enter simulates.

⚙ rule_config thresholds

1 **Normal transaction** — expect: no alert

2 **Amount z-score** — expect: high amount

3 **Impossible travel** — expect: travel

4 **High-velocity burst** — expect: velocity

Amount z-score

expect: high amount

A sudden ₹95,000 electronics purchase on an account that averages about ₹1,180.

Amount ₹95,000 · Minutes since previous txn 245 min

Time of day 14:00 · Payments in the last window 1

Location: Hyderabad · Merchant history: Chai Point · 210 txns, 0 alerts

☐ Unrecognised device (not in account_devices)

Live verdict (updates as you drag)

📈 Amount z-score · FIRES · 🌙 Off-hours · 📱 New device · ⚡ Frequency velocity · ✈ Impossible travel · 🏪 Merchant risk

Where did the card move?

same city



Staying in one place: the speed rule has nothing to measure.

↻ Simulate again · speed: normal

☐ step by step

2 · Inside the engine

Five stages, all inside one INSERT. Nothing leaves the database server.

01 INSERT (client) · 02 Trigger (AFTER INSERT) · 03 Rules (score_transaction) · 04 Score (decision) · 05 Alert (alerts row)

01 Ingestion · a client writes one row

```
INSERT INTO transactions
  (account_id, merchant_id, device_id, location_id, amount, status)
VALUES (7, 14, 12, 3, 95000.00, 'success');   -- Chai Point, Hyderabad
```

1 row stored · txn_id 12068 · amount ₹95,000 · city Hyderabad (stored; scoring has not run yet)

02 Trigger activation · the engine, not the client, makes the next call

```
CREATE TRIGGER trg_after_txn_insert
AFTER INSERT ON transactions FOR EACH ROW
BEGIN
  CALL score_transaction(NEW.txn_id);   -- NEW.txn_id = 12068
END;
```

⚡ trg_after_txn_insert — AFTER INSERT · FOR EACH ROW

03 Rule evaluation · score_transaction() checks six rules against rule_config

📈 Amount z-score — FIRES +1	🌙 Off-hours — clear	📱 New device — clear
n=65 · avg ₹2,623 · sd ₹11,659 → z = 7.92 (limit 3)	14:00 is 7.6% of history (limit 2.0%)	device found in account_devices
⚡ Frequency velocity — clear	✈ Impossible travel — clear	🏪 Merchant risk — SKIPPED
1 txn in 5 min incl. this one (limit 5)	Hyderabad → Hyderabad: 0.0 km in 245 min = 0 km/h (limit 900)	Chai Point has 0 alerts; needs ≥ 3 before it is judged

04 Score and decision · each hit adds 1; any score above 0 raises an alert

score 1 > 0 → RAISE ALERT

Amount z-score. There is no combined threshold: each rule has its own, and one hit is enough.

```
INSERT INTO alerts (txn_id, rule_name, risk_score, reasons, status)
```

Figure C.2 Amount z-score persona.

Sentinel.

A fraud engine that lives *inside* the database. Every `INSERT` is scored by a trigger before the statement returns, so no writer can skip the check.

12,067	13	6	≈2 ms	102
seeded transactions	tables, 3NF	fraud rules	scoring cost per insert	SQL assertions passing

Simulator Engine pipeline Architecture

1 · Choose a transaction

Pick a persona, then bend it: every slider re-evaluates the six rules instantly. Keys 1–4 pick a persona, Enter simulates.

⚙ rule_config thresholds

1	2	3	4
Normal transaction	**Amount z-score**	**Impossible travel**	**High-velocity burst**
expect: no alert	expect: high amount	expect: travel	expect: velocity

Impossible travel

expect: travel

A payment in Tokyo 15 minutes after one in Hyderabad. About 6,300 km in a quarter of an hour.

Amount ₹1,800 — Minutes since previous txn 15 min

Time of day 11:00 — Payments in the last window 2

Location: Tokyo — Merchant history: Chai Point · 210 txns, 0 alerts

☐ Unrecognised device (not in account_devices)

Live verdict (updates as you drag)

📈 Amount z-score · 🌙 Off-hours · 📱 New device · ⚡ Frequency velocity · ✈️ Impossible travel · FIRES · 🏪 Merchant risk

Where did the card move?

6311 km



Implied speed 25,246 km/h vs limit 900

↻ Simulate again speed normal

☐ step by step

2 · Inside the engine

Five stages, all inside one INSERT. Nothing leaves the database server.

01 INSERT (client) · 02 Trigger (AFTER INSERT) · 03 Rules (score_transaction) · 04 Score (decision) · 05 Alert (alerts row)

01 Ingestion · a client writes one row

```
INSERT INTO transactions
  (account_id, merchant_id, device_id, location_id, amount, status)
VALUES (7, 14, 12, 9, 1800.00, 'success');   -- Chai Point, Tokyo
```

1 row stored txn_id 12068 amount ₹1,800 city Tokyo (stored; scoring has not run yet)

window: ● ● 2 payments in 5 min

02 Trigger activation · the engine, not the client, makes the next call

```
CREATE TRIGGER trg_after_txn_insert
AFTER INSERT ON transactions FOR EACH ROW
BEGIN
  CALL score_transaction(NEW.txn_id);   -- NEW.txn_id = 12068
END;
```

⚡ trg_after_txn_insert
AFTER INSERT · FOR EACH ROW

03 Rule evaluation · score_transaction() checks six rules against rule_config

📈 Amount z-score — clear n=65 · avg ₹1,190 · sd ₹718 → z = 0.85 (limit 3)	🌙 Off-hours — clear 11:00 is 8.0% of history (limit 2.0%)	📱 New device — clear device found in account_devices
⚡ Frequency velocity — clear 2 txns in 5 min incl. this one (limit 5)	✈️ Impossible travel — FIRES +1 Hyderabad → Tokyo: 6311.4 km in 15 min = 25,246 km/h (limit 900)	🏪 Merchant risk — SKIPPED Chai Point has 0 alerts; needs ≥ 3 before it is judged

04 Score and decision · each hit adds 1; any score above 0 raises an alert

score 1 > 0 → RAISE ALERT

Impossible travel. There is no combined threshold: each rule has its own, and one hit is enough.

Figure C.3 Impossible-travel persona.

Sentinel.

A fraud engine that lives *inside* the database. Every `INSERT` is scored by a trigger before the statement returns, so no writer can skip the check.

12,067	13	6	≈2 ms	102
seeded transactions	tables, 3NF	fraud rules	scoring cost per insert	SQL assertions passing

Simulator Engine pipeline Architecture

1 · Choose a transaction

Pick a persona, then bend it: every slider re-evaluates the six rules instantly. Keys 1–4 pick a persona, `Enter` simulates.

⚙ rule_config thresholds

1	2	3	4
Normal transaction	**Amount z-score**	**Impossible travel**	**High-velocity burst**
expect: no alert	expect: high amount	expect: travel	expect: velocity

High-velocity burst

expect: velocity

Six payments inside two minutes at different merchants. The sixth insert tips the rule.

Amount ₹905 — Minutes since previous txn 1 min

Time of day 16:00 — Payments in the last window 6

Location: Hyderabad — Merchant history: Chai Point · 210 txns, 0 alerts

☐ Unrecognised device (not in account_devices)

Live verdict (updates as you drag)

📈 Amount z-score · 🌙 Off-hours · 📱 New device · ⚡ Frequency velocity · FIRES · ✈️ Impossible travel · 🏪 Merchant risk

Where did the card move?

same city



Staying in one place: the speed rule has nothing to measure.

↻ Simulate again speed: normal

☐ step by step

2 · Inside the engine

Five stages, all inside one `INSERT`. Nothing leaves the database server.

01 INSERT (client) → 02 Trigger (AFTER INSERT) → 03 Rules (score_transaction) → 04 Score (decision) → 05 Alert (alerts row)

01 Ingestion · a client writes one row

```
INSERT INTO transactions
  (account_id, merchant_id, device_id, location_id, amount, status)
VALUES (7, 14, 12, 3, 905.00, 'success');   -- Chai Point, Hyderabad
```

1 row stored txn_id 12068 amount ₹905 city Hyderabad (stored; scoring has not run yet)

window: ● ● ● ● ● ● 6 payments in 5 min

02 Trigger activation · the engine, not the client, makes the next call

```
CREATE TRIGGER trg_after_txn_insert
AFTER INSERT ON transactions FOR EACH ROW
BEGIN
  CALL score_transaction(NEW.txn_id);   -- NEW.txn_id = 12068
END;
```

⚡ trg_after_txn_insert
AFTER INSERT · FOR EACH ROW

03 Rule evaluation · score_transaction() checks six rules against rule_config

📈 Amount z-score — clear n=65 · avg ₹1,176 · sd ₹715 → z = -0.38 (limit 3)	🌙 Off-hours — clear 16:00 is 7.1% of history (limit 2.0%)	📱 New device — clear device found in account_devices
⚡ Frequency velocity — FIRES +1 6 txns in 5 min incl. this one (limit 5)	✈️ Impossible travel — clear Hyderabad → Hyderabad: 0.0 km in 1 min = 0 km/h (limit 900)	🏪 Merchant risk — SKIPPED Chai Point has 0 alerts; needs ≥ 3 before it is judged

04 Score and decision · each hit adds 1; any score above 0 raises an alert

score 1 > 0 → RAISE ALERT

Frequency velocity. There is no combined threshold: each rule has its own, and one hit is enough.

Figure C.4 High-velocity persona.